

Name: _____ Tutorial group: _____

Matriculation number:

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31 October 2022 MH1812 Test 2 60 minutes

QUESTION 1. (20 marks)

Solve the linear recurrence relation $a_n = 6a_{n-1} - 9a_{n-2}$ with initial values $a_0 = 1, a_1 = 9$.

[Solution:] The characteristic equation is $x^2 - 6x + 9 = 0$, which has a repeated root $x_1 = x_2 = 3$. Thus the general form is $a_n = c_1 3^n + c_2 n 3^n$. Plugging in $a_0 = 1$ and $a_1 = 9$, we have

$$\begin{aligned} c_1 &= 1 \\ 3c_1 + 3c_2 &= 9 \end{aligned}$$

Thus $c_1 = 1$ and $c_2 = 2$. The solution of the recurrence relation is therefore

$$a_n = 3^n + 2n3^n = 3^n(2n + 1).$$

QUESTION 2. (20 marks)

There are 3 different textbooks.

(a) (10 points) How many ways can we choose 10 books in total if each textbook has 10 identical copies?

[Solution:] Assume there are textbooks A, B, C, and we choose x_1 copies of A, x_2 copies of B, and x_3 copies of C, we have

$$x_1 + x_2 + x_3 = 10,$$

where $0 \leq x_1, x_2, x_3 \leq 10$. This is the same as $x_1, x_2, x_3 \geq 0$ for this problem, which corresponds to the multiset combination with $n = 10, k = 3$. Thus, there are $\binom{n+k-1}{k-1} = \binom{12}{2} = 66$ ways.

(b) (10 points) How many ways can we choose 10 books in total if each textbook has 9 identical copies?

[Solution:] Similar to above, except now $0 \leq x_1, x_2, x_3 \leq 9$, i.e., the 10 books should not be of the same type, and there are 3 cases for all books to be of the same type, hence $\binom{12}{2} - 3 = 63$.

For graders only	Question	1	2(a)	2(b)	3(a)	3(b)	3(c)	4(a)	4(c)	Bonus	Total
	Marks										

QUESTION 3. (30 marks)

Give the set $A = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, and a relation R defined on A as for all $x, y \in A, (x, y) \in R$ if and only if $x \mid y$.

(a) (10 points) Express R explicitly by listing out all its elements.

[Solution:]
 $R = \{(1, 1), (1, 2), (1, 3), (1, 4), (1, 5), (1, 6), (1, 7), (1, 8), (1, 9),$
 $(2, 2), (2, 4), (2, 6), (2, 8), (3, 3), (3, 6), (3, 9), (4, 4), (4, 8),$
 $(5, 5), (6, 6), (7, 7), (8, 8), (9, 9)\}.$

(b) (12 points) Is R Reflexive? Symmetric? Anti-symmetric? Transitive? an Equivalence Relation ? a Partial Order ?(Justify your answer)

[Solution:]

- R is reflexive as it holds that $x \mid x$ for all $x \in A$.
- R is not symmetric because $(2, 4) \in R$ but $(4, 2) \notin R$.
- R is anti-symmetric. If $x \mid y$ and $y \mid x$ both hold for positive integers x, y , then $y = kx$ and $x = \ell y$ for some integers $k, \ell \geq 1$, we have $x = k\ell x$, which means that $k\ell = 1$ and thus $k = \ell = 1$. This implies that $x = y$.
- R is transitive. If $x \mid y$ and $y \mid z$, then $y = kx$ and $z = \ell y$ for integers k, ℓ , so $z = k\ell x$, or $x \mid z$.
- R is not an equivalence relation because it is not symmetric.
- R is a partial order because it is reflexive, anti-symmetric and transitive.

(c) (8 points) Find R^2 .

[Solution:] Since R is reflexive and transitive, it holds that $R^2 = R$.

QUESTION 4.**(30 marks)**(a) (20 points) Prove or disprove, using membership table that for any sets A, B, C

$$(A - B) \cap C = (A \cap C) - (B \cap C)$$

[Solution:]

A	B	C	$A - B$	$(A - B) \cap C$	$A \cap C$	$B \cap C$	$(A \cap C) - (B \cap C)$
0	0	0	0	0	0	0	0
0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0
0	1	1	0	0	0	1	0
1	0	0	1	0	0	0	0
1	0	1	1	1	1	0	1
1	1	0	0	0	0	0	0
1	1	1	0	0	1	1	0

It is obvious to see that the column for $(A - B) \cap C$ and for $(A \cap C) - (B \cap C)$ are identical, hence equality holds.

(b) (10 points) Given a set A with $|A| = n$ where n is a positive integer, how many possible anti-symmetric relations in total can be defined on A ?[Solution:] The answer is $2^n 3^{n(n-1)/2}$.

The 2^n factor corresponds to whether or not to include each pair (x, x) in the relation. The $3^{\binom{n}{2}}$ factor corresponds to three cases for each pair (x, y) with $x \neq y$: (x, y) is included in the relation, (y, x) is included in the relation, or neither.

BONUS QUESTION.**(10 marks)**[Points will be given to *fully* correct solutions only. The total mark of this test is capped at 100 marks.]Let $N = (2 + \sqrt{3})^{2022}$. What is the first digit at the left of the decimal point of N ?

[Solution:] Consider

$$a_n = (2 + \sqrt{3})^n + (2 - \sqrt{3})^n$$

then a_n satisfies the recurrence relation (one can verify using the characteristic equation)

$$a_n = 4a_{n-1} - a_{n-2}$$

with $a_0 = 2$ and $a_1 = 4$.

Using (strong) mathematical induction, one can obtain that

$$a_n \equiv \begin{cases} 2, & n = 3k \\ 4, & n = 3k + 1 \\ 4, & n = 3k + 2 \end{cases} \pmod{10}.$$

Therefore, $a_{2022} = (2 + \sqrt{3})^{2022} + (2 - \sqrt{3})^{2022}$ ends with 2. Clearly $(2 - \sqrt{3})^{2022} < 2 - \sqrt{3} < 1$. Hence, the integer part of $(2 + \sqrt{3})^{2022}$ ends with 1.

[Alternative solution:] We obtain $a_n = 4a_{n-1} - a_{n-2}$ as above. Then

$$\begin{aligned} a_n &\equiv 4a_{n-1} - a_{n-2} \\ &\equiv 4(4a_{n-2} - a_{n-3}) - a_{n-2} \\ &\equiv 15a_{n-2} - 4a_{n-3} \\ &\equiv 5a_{n-2} - 4a_{n-3} \\ &\equiv 5(4a_{n-3} - a_{n-4}) - 4a_{n-3} \\ &\equiv -5a_{n-4} - 4a_{n-3} \\ &\equiv -5a_{n-4} - 4(4a_{n-4} - a_{n-5}) \\ &\equiv -a_{n-4} + 4a_{n-5} \\ &\equiv -(4a_{n-5} - a_{n-6}) + 4a_{n-5} \\ &\equiv a_{n-6} \pmod{10} \end{aligned}$$

Then $a_{2022} \equiv a_0 \equiv 2 \pmod{10}$. The rest is the same as above.